

Types of Microscopes

TYPE	DESCRIPTION	KEY APPLICATIONS IN ELECTRONICS	MAGNIFICATION RANGE (TYPICAL)	ADVANTAGES	LIMITATIONS
Stereo Microscope	Uses two optical paths for 3D visualisation with a wide field of view	Inspecting PCBs, solder joints, wire bonding, and component alignment during assembly and rework	5x–80x	• 3D depth perception for soldering/rework • Easy to use, wide field view • Compatible with ESD-safe benches	• Limited magnification • Not suitable for nano-scale inspection
Compound Microscope	High-magnification light microscope with multiple objective lenses for 2D imaging	Examining fine circuit traces, contamination, and micro-cracks in semiconductors	40x–1000x	• High resolution imaging • Good for fine details in thin films and defects	• Flat 2D view (no depth) • Requires thin/transparent samples
Inverted Microscope	Views sample from below; suitable for larger/mounted specimens	Inspecting encapsulated PCBs, coated wafers, or layered semiconductor processes	40x–400x	• Can handle bulky/mounted samples • Useful for live processes or coated layers	• Moderate magnification • Bulkier and costlier than stereo microscopes
Metallurgical Microscope	Uses reflected light for opaque/metallic surfaces	Inspecting metal interconnects, solder balls, welds, wafers for cracks or impurities	50x–500x	• Excellent for opaque and reflective surfaces • Identifies cracks, voids, inclusions in metals	• Limited to surface inspection • Less effective for transparent samples
Polarising Microscope	Uses polarised light to enhance contrast in birefringent materials	Studying stress patterns, crystalline defects, thin film alignment	10x–400x	• Highlights stress and crystalline structures • Useful for materials R&D	• Niche application • Requires specialised training
Digital Microscope	Integrates optics, camera, and display; may include software for measurement and reporting	Documentation, training, defect reporting, and high-volume inspections in production lines	20x–500x (variable zoom)	• Easy sharing and recording • Software-enabled measurement • Useful for remote collaboration	• Lower optical quality than dedicated optical scopes • May have lag or resolution limits
Scanning Electron Microscope (SEM)	Uses electron beams for ultra-high-resolution (nano-scale) imaging	Failure analysis, nano-defect detection, material studies, wafer surface morphology	Up to 100,000x+	• Extremely high resolution • Enables material composition analysis (with EDS add-on) • Critical for advanced semiconductor R&D	• Very expensive • Requires vacuum and skilled operators • Slow compared to optical scopes
Atomic Force Microscope (AFM)	Uses a sharp probe to scan the surface at the atomic level, measuring forces between the tip and sample	Nano-scale surface characterisation, thin film measurement, defect detection in semiconductors, and 3D surface profiling	Up to ~1,000,000x (atomic resolution)	• Provides true 3D topographic images • Can measure mechanical, electrical, and magnetic properties at nano-scale • Works in air, vacuum, or liquid environments	• Very slow scan speed • Limited scan area • Delicate probes and high operator skill required • Very high cost

KEY IPC STANDARDS

IPC standards ensure visual inspections meet quality benchmarks, with microscope usage tied to magnification and resolution requirements. Key standards include:

- **IPC-A-610 (Acceptability of Electronic Assemblies):** Defines visual criteria for assemblies, requiring magnification aids (e.g., 1.75x–4x for general inspection, up to 10x for fine details). It influences microscope selection—digital systems for documentation, binocular for precise work—and mandates lighting and training.
- **IPC-J-STD-001 (Requirements for Soldered Electrical and Electronic Assemblies):** Specifies inspection methods, including microscope use for solder joint evaluation
- **IPC-A-600 (Acceptability of Printed Boards):** Focuses on PCB appearance and structure, recommending microscopes for internal inspections.
- **IPC-TM-650 (Test Methods Manual):** Provides guidelines for evaluating PCBs, including visual methods under magnification.

For full IPC standards, visit ipc.org

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